

# RORA: Robust Free-Text Rationale Evaluation

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## Abstract

Free-text rationales play a pivotal role in explainable NLP, bridging the knowledge and reasoning gaps behind a model’s decision-making. However, due to the diversity of potential reasoning paths and a corresponding lack of definitive ground truth, their evaluation remains a challenge. Existing evaluation metrics rely on the degree to which a rationale *supports* a target label, but we find these fall short in evaluating rationales that inadvertently *leak the labels*. To address this problem, we propose RORA, a RObust free-text RAtionale evaluation against label leakage.<sup>1</sup> RORA quantifies the new information supplied by a rationale to justify the label. This is achieved by assessing the conditional  $\mathcal{V}$ -information (Hewitt et al., 2021) with a predictive family robust against leaky features that can be exploited by a small model. RORA consistently outperforms existing approaches in evaluating human-written, synthetic, or model-generated rationales, particularly demonstrating robustness against label leakage. We also show that RORA aligns well with human judgment, providing a more reliable and accurate measurement across diverse free-text rationales.

## 1 Introduction

The ability of large language models (LLMs) to generate free-text rationales that elaborate on their decision-making processes holds promise for explainable NLP, either in the form of a reasoning chain (Wei et al., 2022; Yao et al., 2023) or post-hoc explanations (Madaan et al., 2023; Zheng et al., 2023). Previous works have also collected human-written rationales to enhance model reasoning and the generation of free-text rationales (Rajani et al., 2019; Camburu et al., 2018; Aggarwal et al., 2021; Geva et al., 2021).

However, evaluating these rationales remains an open problem because of the diversity of reasoning

paths and the lack of definitive ground truth (Chan et al., 2022). As a result, existing metrics rely on measuring how much the rationale supports a given label. This is usually achieved by comparing predictions of models trained with and without rationales. For example, Leakage-Adjusted Simulatability (LAS) (Hase et al., 2020) and Rationale Quality (RQ) (Wiegrefe et al., 2020) measure rationale quality through the difference in accuracy. Alternatively, Rationale Evaluation with conditional- $\mathcal{V}$ -information (REV) (Chen et al., 2023b) evaluates the reduction in model predictive uncertainty upon conditioning on the rationale.

Yet all these methods are vulnerable to *label leakage* (Li et al., 2022; Chen et al., 2023c; Ludan et al., 2023): the rationale inadvertently *paraphrasing* or *restating* labels, creating a spurious shortcut (Geirhos et al., 2020) for the evaluation model to infer the label. The critical issue stems from the mismatch in objectives: existing methods evaluate how easy it is to utilize information in the rationale, but rationales are *explanations*, whose quality does not always come with simplicity. The best explanation has to support the answer through some sense of mechanisms, such as methodically considering a set of axioms and running through a deductive chain (Bechtel and Abrahamsen, 2005; Keil, 2006; Glennan, 2002), without which they are mere “effects” (Cummins, 2000). Figure 1 shows an example where existing evaluation methods are highly sensitive to label leakages in paraphrased or restated rationales, while in fact, these label leakages merely increase the predictive probability without providing any meaningful explanations.<sup>2</sup>

With this objective in mind, we introduce RORA, a novel approach to evaluate rationales robust to label leakage. RORA’s construction consists of

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<sup>1</sup><https://github.com/zipJiang/RORA>

<sup>2</sup>Note that scores between different evaluation metrics are not directly comparable because of different scales and criteria. In this paper, our analysis mainly focuses on the ranking and relative differences within each metric.

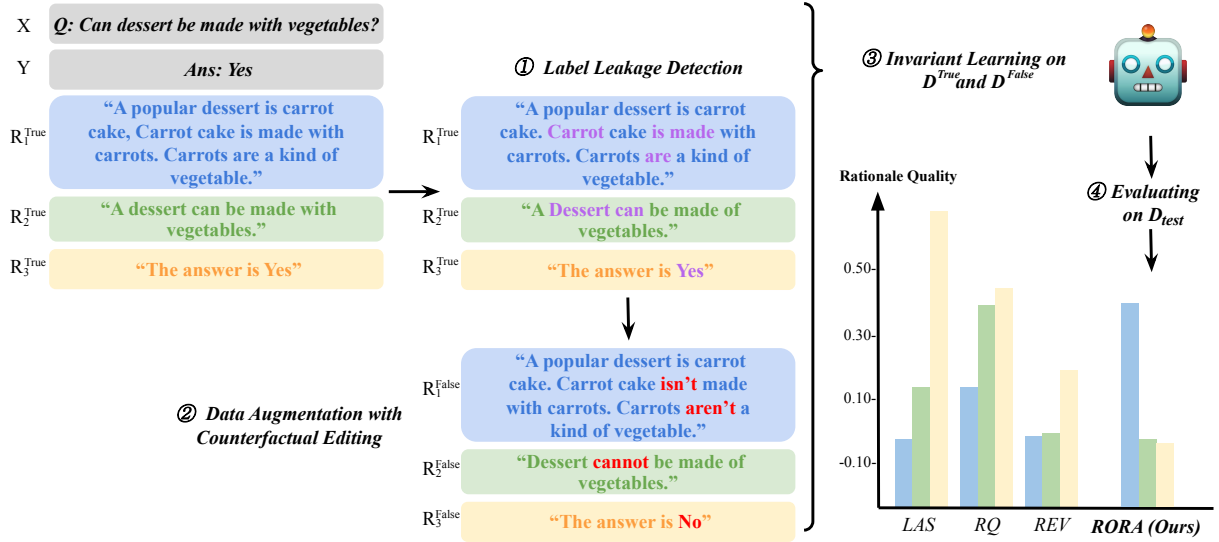


Figure 1: RORA framework for evaluating rationales  $R_1^{\text{True}}$ ,  $R_2^{\text{True}}$ ,  $R_3^{\text{True}}$ . Existing baselines are highly sensitive to rationales that **simply restate the label** or **paraphrase the given question and label**, leading to inflated scores compared to the **human-annotated** rationale. In contrast, RORA provides an informativeness score that better characterizes rationale quality. It is achieved by ① detecting potential leakage tokens in the rationale (§3.1) and ② generate additional training data with counterfactual editing for data augmentation (§3.2), followed by ③ training an evaluation model invariant to label leakage (§3.3).

three stages as illustrated in Figure 1. First, we fit a small model and identify label-leaking tokens via its gradient-based attributions (§3.2). After that, we generate additional training data with counterfactual editing (Ross et al., 2021) (§3.2). Finally, we force the evaluation model to ignore these label-leaking tokens through invariant learning (Arjovsky et al., 2020) (§3.3). Our approach aligns with the human perception that explanations should apparently increase the understanding of a given phenomenon by helping to create knowledge and to develop better theories (Wilson and Keil, 1998). On the contrary, label leakage tends to be repetitive and tautological (Aslanov and Guerra, 2023), dominating the insightful parts of the explanation.

We compare RORA with baseline metrics (Hase et al., 2020; Wiegrefe et al., 2020; Chen et al., 2023b) in evaluating various synthetic and human-annotated rationales, with or without label leakage, on three QA datasets. RORA consistently outperforms baseline metrics by providing robust evaluations against label leakages. We also compare RORA against model-generated rationale evaluation and demonstrate its better agreement with human evaluation.

## 2 Motivation and Backgrounds

Following previous work by Chen et al. (2023b), RORA adopts the framework of conditional  $\mathcal{V}$ -

information (Hewitt et al., 2021). In this section, We outline the general idea of this framework and point out how the objective of RORA is critically different from previous attempts.

### 2.1 Conditional $\mathcal{V}$ -information

The theory of  $\mathcal{V}$ -information, as proposed by Xu et al. (2020), focuses on quantifying usable information under computational constraints. Specifically, it examines how much usable information about a random variable  $Y$  can be derived from another random variable  $X$  by applying functions belonging to a specified set known as the *predictive family*  $\mathcal{V}$ . Due to its transparent connection to baseline probing, Hewitt et al. (2021) propose to extend  $\mathcal{V}$ -information to multivariable cases as an evaluation of information beyond a baseline.

**Definition 1** (Multivariable  $\mathcal{V}$ -information). Let  $X_1, X_2, \dots, X_n \in \mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_n$  and  $Y \in \mathcal{Y}$  be random variables. Let  $\mathcal{V}$  be a multivariable predictive family.<sup>3</sup> Then the conditional multivariable  $\mathcal{V}$ -information from  $X_j$  to  $Y$ , where  $j \in \{1, 2, \dots, n\}$ , conditioned on prior knowledge

<sup>3</sup>We refer the readers to the work of Xu et al. (2020) We adopt their notation (rather than that of Hewitt et al. (2021)) since it abstracts away actual implementation from the underlying idea.

$C \subset \{X_1, \dots, X_n\}$ , is defined as

$$I_{\mathcal{V}}(X_l \rightarrow Y|C) = H_{\mathcal{V}}(Y|C) - H_{\mathcal{V}}(Y|C \cup X_l), \quad (1)$$

where

$$H_{\mathcal{V}}(Y|C) = \inf_{f \in \mathcal{V}} \mathbb{E}_{c,y} \left[ -\log f[c, \{\emptyset, \dots, \emptyset\}](y) \right]. \quad (2)$$

is the multivariable predictive  $\mathcal{V}$ -entropy.<sup>4</sup>

*Remark.* Equation 1 is also called the “conditional  $\mathcal{V}$ -information” (and Equation 2 “conditional  $\mathcal{V}$ -entropy”). Equation 1 characterizes how much uncertainty of a variable  $Y$  can be reduced when a certain class of function was used to extract knowledge from observing  $X_l$ . The formulation of Equation 1 allows estimation of mutual information via optimization over  $\mathcal{V}$ , i.e., through gradient descent. It is important to note that the definition of the predictive family ensures that a predictive family can ignore any feature subset while the remaining subset  $c$  still has the same output. In our case, it ensures that one can always construct a valid, more limited predictive family by forcing ignorance on leaky features.

## 2.2 Evaluating Label-Leaking Rationales

The conditional  $\mathcal{V}$ -information provides a natural way to evaluate the informativeness of rationales. For a natural language problem with input  $X$  and label  $Y$ , A direct application of multivariable  $\mathcal{V}$ -information to evaluate the quality of rationale  $R$  is

$$Q(R) = I_{\mathcal{V}}(R \rightarrow Y|X). \quad (3)$$

Yet, as Hase et al. (2020) pointed out, this approach fails to distinguish between different ways a rationale provides knowledge for a model’s prediction. For example, a bad rationale can obtain a very high  $Q(R)$  by restating the label to be predicted. To solve this problem, REV (Chen et al., 2023b) proposes to introduce a vacuous rationale  $B$  that is simply a declarative combination of  $X$  and  $Y$

$$\text{REV}(R) = I_{\mathcal{V}}(R \rightarrow Y|B). \quad (4)$$

However, this requires that  $Y$  is not completely determined by  $B$ , which can be easily violated by

<sup>4</sup>Following Xu et al. (2020)  $\emptyset$  is used to indicate that no information is provided for this feature. In practice, it is common to mask out this feature with constant values (Hewitt et al., 2021).

label leakage in  $B$  that makes  $H_{\mathcal{V}}(Y|B)$  close to 0.<sup>5</sup>

Under a similar framework where conditional  $\mathcal{V}$ -entropy  $H_{\mathcal{V}}$  in Equation 1 is replaced by prediction accuracy, other methods are proposed to mitigate the impact of label leakage. Hase et al. (2020) propose LAS: an averaged accuracy between leaking and non-leaking rationales which are classified by whether a simulator model can predict  $Y$  solely from the explanation  $R$ . However, the prevalence of label leakage in free-text rationales may overwrite the effect of macro-averaging. The metric is simply undefined when all rationales in the dataset leak labels (Pruthi et al., 2022). Sia et al. (2022) suggest that using logical counterfactuals may circumvent the label leakage issue. However, their approach does not generalize to most NLP tasks because rationales in free-text format are usually too complex to be parsed logically.

Suppose a free-text rationale  $R$  consists of label leakage  $R_L$  and non-leaky information  $R_R$  (i.e.,  $R = R_L \cup R_R$ ). We propose that a good evaluator  $\mathcal{S}$  of the informativeness of rationale  $R$  would be

$$\begin{aligned} \mathcal{S}(R) &= I_{\mathcal{V}}(R_R \rightarrow Y|X), \\ &= I_{\mathcal{V}}(R \setminus R_L \rightarrow Y|X). \end{aligned} \quad (5)$$

Notice that we do not condition on  $B$ , as in our formulation, vacuous rationales are similar to spurious features about which our evaluator should be ignorant (Xu et al., 2020).  $R_L$  and  $R_R$  are usually deeply entangled in natural language utterances, and it is difficult to isolate  $R_R$  completely. To address this problem, we propose RORA to approximate Equation 5 by identifying and forcing the model to unlearn  $R_L$ , as detailed in §3.

## 3 RORA for Rationale Evaluation

We describe our framework (Figure 1) to estimate Equation 5 as a measurement of the informativeness of rationales.

### 3.1 Leakage Detection

To detect leaking features  $R_L$ , we fit a relatively small model on corresponding rationales  $\Phi_{\text{small}} : \mathcal{R} \rightarrow \mathcal{P}(\mathcal{Y})$  that predicts the label distribution from the rationale alone. We then calculate the attribution of its prediction to each input token using Integrated Gradient (Sundararajan et al., 2017).

<sup>5</sup>Conditioning on  $B$  may cause other problems, of which we refer the readers to Appendix B for a more detailed discussion.

This token-level attribution is then averaged over the whole dataset to get global attribution for each token. Figure 1 shows example attribution on different rationales. The highlight (in purple) denotes the token is critical to the model’s prediction as its attribution is above a given threshold.

We would prefer the model to be *less contextualized, less-trained, and smaller in size*. Our motivation is that smaller models won’t be able to learn reasoning features, so their attributions tend to rely on shortcuts (Geirhos et al., 2020) like label leakage. In addition, attribution for smaller models is known to be more reliable and meaningful (Neely et al., 2022).

### 3.2 Data Augmentation

Given the original dataset  $D = (X, R, Y)$ , we generate partially counterfactual datasets

$$D^e = (X, R^e, Y), \forall e \in \mathcal{Y},$$

where editing is constrained to the spans identified in the previous step. Following MICE (Ross et al., 2021), we train a label-associated sequence to sequence infiller. Specifically for the  $i$ -th datapoint  $(x_i, r_i, y_i) \in D$ , we create training data by processing the sentence  $y_i \oplus x_i \oplus r_i$  with the same denoising objective as in T5 pretraining (Raffel et al., 2020), where a single sentinel mask replaces each consecutive span from gradient attribution (§3.1) as inputs and the training objective is to predict masked spans. Notice that we only mask within the rationale to control artificially introduced artifacts as discussed in the Appendix A. To augment a dataset  $D$ , we decode infilling for these masked spans for each  $e \in \mathcal{Y}$  to create  $D^e$ . Notice that, unlike usual counterfactual generation, we do not necessarily require the generated rationale  $r_i^e$  with prepending label  $e$  to be highly associated with that label. In fact, the training of the editing model in MICE can be thought of as estimating

$$I_{\mathcal{V}}(Y \rightarrow R|X).$$

Therefore, only those parts of the rationale that are easy to predict from the label will be edited, and most other spans will likely remain the same (as shown in Figure 1), as the model predicts these spans mostly from the static  $x_i$  instead of the flipped  $y$ . This is a desirable property in our case, as it leads to hyperparameters that are easy to tune, with which we apply a very aggressive removal threshold while still preserving most of the  $R_R$ .

### 3.3 Computing RORA

Given all the augmented datasets  $\mathcal{D} := \{D^e | \forall e \in \mathcal{Y}\}$ , we train our evaluator on the joint of these datasets.<sup>6</sup> We learn an invariant predictor parameterized by  $\Phi$  by applying the Invariant Risk Minimization (IRMv1) regularization (Arjovsky et al., 2020)

$$\begin{aligned} \mathcal{L}(\Phi) = & \sum_{D^e \in \mathcal{D}} \mathcal{L}^e(\Phi) \\ & + \lambda \|\nabla_{w|w=1.0} \mathcal{L}^e(w \cdot \Phi)\|^2, \end{aligned} \quad (6)$$

where  $\mathcal{L}^e$  is the Empirical Risk Minimization (ERM) term on dataset  $D^e$  and the second term is the IRM regularization to enforce  $\Phi$  to be simultaneously optimal on all environments  $D^e$ . In our case, it is either the factual data or the counterfactual augmentation.  $\lambda$  is a hyperparameter balancing the tradeoff between the predictive power (an ERM term) and the invariance of the predictor across different environments. When the underlying model is a classifier, Equation 6 is directly applicable with normal cross-entropy loss. However, considering the increasing trend of directly decoding label sequence from pre-trained language models like T5 (Raffel et al., 2020; Nogueira et al., 2021) and to better align with previous works (Chen et al., 2023b), we use the following variant with label decoding:

$$\begin{aligned} \mathcal{L}(\Phi) = & \sum_{D^e \in \mathcal{D}} \mathbb{E}_{(x,r,y) \sim D^e} [-\log p_{\Phi}(y|x, r)] \\ & + \lambda \|\nabla_{w|w=1.0} \log \frac{p_{\Phi}(y|x, r)}{\sum_{y' \in \mathcal{Y}} p_{\Phi}(y'|x, r)}\|^2. \end{aligned} \quad (7)$$

Notice that we still train to maximize probability over all possible strings in the ERM term to facilitate greedy decoding. In an ideal world with infinite data, this is equivalent to the conditional probability on the set of all possible labels.

Additionally, we train a baseline model  $\Theta$  by minimizing

$$\mathcal{L}_{\text{Base}}(\Theta) = \mathbb{E}_{(x,r,y) \sim D} [-\log p_{\Theta}(y|x, \emptyset)]. \quad (8)$$

As Equation 5 can be decomposed into the difference between two conditional  $\mathcal{V}$ -entropy terms

$$S(R) = H_{\mathcal{V}}(Y|X) - H_{\mathcal{V}}(Y|R \setminus R_L \cup X).$$

<sup>6</sup>In reality, it is possible to reuse the original dataset  $D$ , as in a single batch the model sees corresponding data points from all datasets.

| Dataset    | Question                         | <i>gold</i>                                                                      | <i>leaky</i>        | <i>gold<sub>leaky</sub></i>                                                                          | <i>vacuous</i>                   |
|------------|----------------------------------|----------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------|----------------------------------|
| StrategyQA | Could Chuck Norris ride a horse? | Chuck Norris is a person. Horses are bigger than people. People can ride horses. | The answer is True. | Chuck Norris is a person. Horses are bigger than people. People can ride horses. The answer is True. | Chuck Norris could ride a horse. |

Table 1: Example of human-annotated and synthetic leaky rationales for fixed-label QA task. *leaky* verbatim leaks the label, *vacuous* is the declarative combination of the question and the given answer, and *gold<sub>leaky</sub>* is simply the concatenation of the *gold* and *leaky* rationales.

With  $\Theta$  and  $\Phi$  from the same predictive family, the RORA score  $S(R)$  can thus be estimated as

$$\hat{S}(R) = \mathbb{E}_{(x,r,y) \sim D_{\text{test}}} [\log p_{\Phi}(y|x, r) - \log p_{\Theta}(y|x, \emptyset)]. \quad (9)$$

Notice that the testing dataset  $D_{\text{test}}$  has the same rationale distribution as the original dataset  $D$ .

If the model is simultaneously optimal for all  $D^e \in \mathcal{D}$  (Arjovsky et al., 2020), it is ignorant (Xu et al., 2020) of the features that are different on  $\forall D^e \in \mathcal{D}$  (otherwise, the predictive distribution will be different while the label distribution is the same). In our case, these features are supposed to be related to label leakages. We also discuss why naively masking out leaking parts does not work for free-text rationale evaluation in Appendix A.

## 4 Experiments and Results

To demonstrate the effectiveness of RORA, we evaluate human-annotated, synthetic, and model-generated rationales on two distinct QA tasks. For synthetic data, we (1) write a sentence that verbatim leaks the label (*leaky*); (2) perturb the human-annotated rationales (*gold*) to include the *leaky* rationale (*gold<sub>leaky</sub>*); (3) create declarative rewrites (*vacuous*) of the question-answer pairs using a question rewriter (Chen et al., 2021).<sup>7</sup> Table 1 shows examples of these rationales.<sup>8</sup> For all synthetic and human-annotated data, **a good metric for rationale quality should be able to evaluate *gold* and *gold<sub>leaky</sub>* reasonably close, while ranking them above *leaky* and *vacuous* rationales.** For model-generated rationales, **a good metric should evaluate rationales aligning with human judgments.** To create model-generated rationales of varying qualities, we choose GPT-4 (OpenAI, 2023), GPT-3.5 (Ouyang et al., 2022), Llama2-7b (Touvron et al., 2023) and Flan-T5 large (Chung

et al., 2022), each being prompted with the same two demonstrations to generate a rationale for all question-answer pairs in StrategyQA (examples shown in Appendix C.4).

Without further specification, we use the gradient attribution (Sundararajan et al., 2017) from Fast-Text model (Joulin et al., 2017) to identify label-leaking tokens to mask. We use T5-base (Raffel et al., 2020) to generate counterfactual data. Finally, two additional T5-base models are trained under Equation 7 and Equation 8 as evaluation models to compute the RORA score using Equation 9.<sup>9</sup>

We compare RORA with three strong baselines: LAS (Hase et al., 2020), RQ (Wiegrefe et al., 2020), and REV (Chen et al., 2023b). All baselines use T5-base as the evaluation model. Note that the boundaries of these metrics differ depending on the choice of the  $H_V$  function in Equation 1. Specifically, LAS and RQ, which utilize accuracy, lie within the  $[-1, 1]$  interval. In contrast, RORA and REV, employing the conditional  $V$ -entropy (Equation 2), have a theoretical boundary of  $[-\infty, \infty]$ . Because of different scales and evaluation criteria, our analysis mainly focuses on the ranking over different rationale types, as well as the relative difference within each metric. We also do an ablation study with (RORA<sub>ablation</sub>) with all the attributions, counterfactual generation and invariant learning removed. This is equivalent to the formulation in Equation 3 in which the evaluator  $\Phi$  is directly trained on the given rationales and questions to maximize the likelihood of labels.

### 4.1 Fixed-Label QA Task

We apply RORA to StrategyQA (Geva et al., 2021), a multi-step commonsense reasoning dataset with fixed label set  $\{\text{True}, \text{False}\}$ . We concatenate the provided list of relevant facts to create *gold* rationales. Notice that *gold* rationale created this way

<sup>7</sup>[https://huggingface.co/domenicrosati/question\\_converter-3b](https://huggingface.co/domenicrosati/question_converter-3b)

<sup>8</sup>For space concerns examples for ECQA are shown in Table 5 in Appendix C

<sup>9</sup>Note that the baseline evaluation model  $\Theta$  does not condition on rationales, thus  $\Theta$  can be trained once and the result of  $\mathbb{E}_{(x,r,y)} \log p_{\Theta}(y|x, \phi)$  applies to the evaluation of various rationales types for the same dataset.

| Rationales →             | Synthetic Leaky |                             |                |              | Model Generated |         |           |               |
|--------------------------|-----------------|-----------------------------|----------------|--------------|-----------------|---------|-----------|---------------|
| Metrics ↓                | <i>gold</i>     | <i>gold<sub>leaky</sub></i> | <i>vacuous</i> | <i>leaky</i> | GPT-4           | GPT-3.5 | Llama2-7B | Flan-T5 Large |
| RoRA                     | 0.115           | 0.119                       | 0.043          | 0.038        | 0.283           | 0.253   | 0.100     | 0.061         |
| RoRA <sub>DeBERTa</sub>  | 0.381           | 0.348                       | 0.026          | 0.024        | 0.474           | 0.459   | 0.232     | 0.132         |
| RoRA <sub>ablation</sub> | 0.121           | 0.673                       | 0.505          | 0.673        | 0.501           | 0.412   | 0.155     | 0.102         |
| REV                      | -0.038          | 0.145                       | -0.005         | 0.147        | -0.066          | -0.083  | -0.035    | -0.013        |
| LAS                      | 0.024           | NaN                         | 0.131          | NaN          | 0.161           | 0.215   | 0.084     | 0.027         |
| RQ                       | 0.138           | 0.406                       | 0.376          | 0.406        | 0.316           | 0.311   | 0.121     | 0.071         |
| Human Eval               | —               | —                           | —              | —            | 2.69            | 2.32    | 1.16      | 0.39          |

Table 2: Evaluation results on StrategyQA (Geva et al., 2021) over synthetic and model-generated rationales. **Both RoRA and RoRA<sub>DeBERTa</sub> rank *gold* and *gold<sub>leaky</sub>* high and close to each other, while they rank *vacuous* and *leaky* close to zero, indicating strong invariance to various label leakages. Additionally, RoRA and RoRA<sub>DeBERTa</sub> consistently align the ranking of model-generated rationales with those of humans.**

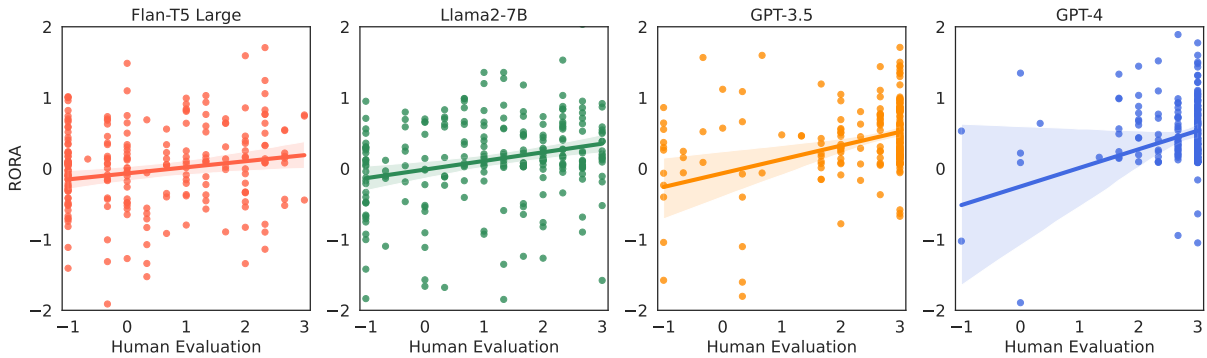


Figure 2: Linear regression on pointwise score correlation between human evaluation results and RORA scores. Shades correspond to a 95% confidence interval.

has all the essential facts but does not explicitly define each reasoning step to be taken.

Table 2 shows evaluation results on human-annotated, synthetic, and model-generated rationales. Evidently, only full pipeline RORA ranks *gold* and *gold<sub>leaky</sub>* the highest, much higher than scores of *vacuous* and *leaky*. This aligns with the human intuition that rationales with new, useful information should be ranked higher while restating labels should not affect scoring. However, all existing baselines violate the intuition and assign much higher scores to three synthetic leaky rationales than the *gold* rationale. We note that LAS is undefined for *leaky* and *gold<sub>leaky</sub>* rationales as it classifies all examples to leaked, thus causing a division-by-zero problem when averaging non-leaked examples, a problem discussed by Pruthi et al. (2022).

We observe similar trends with RoRA<sub>DeBERTa</sub>, which is a variation of RoRA with the final two T5-base evaluation models replaced by DeBERTaV3-large (He et al., 2023) trained with Equation 6

directly. This indicates that RoRA is model-agnostic, demonstrating comparable effectiveness when using different baseline models. On the other hand, RoRA<sub>ablation</sub> is unable to provide comparable scores to *gold* and *gold<sub>leaky</sub>*, yet it assigns the highest to *leaky*, which again highlights the effectiveness of the RoRA approach in mitigating label leakage during evaluation.

**Human Evaluation** We conduct a human evaluation for free-text rationales on Amazon Mechanical Turk. We annotate all 229 instances in our custom test split of StrategyQA. For each instance, we annotate rationales generated by all four models. For each rationale, the annotator is first asked to provide a binary judgment on whether the rationale supports the target label. If the answer is positive, the annotators are asked to judge how much new information the rationale provides beyond restating the question and the answer. This yields a natural 5-point scale to which we give scores in  $\{-1, 0, 1, 2, 3\}$  respectively. Each rationale is an-

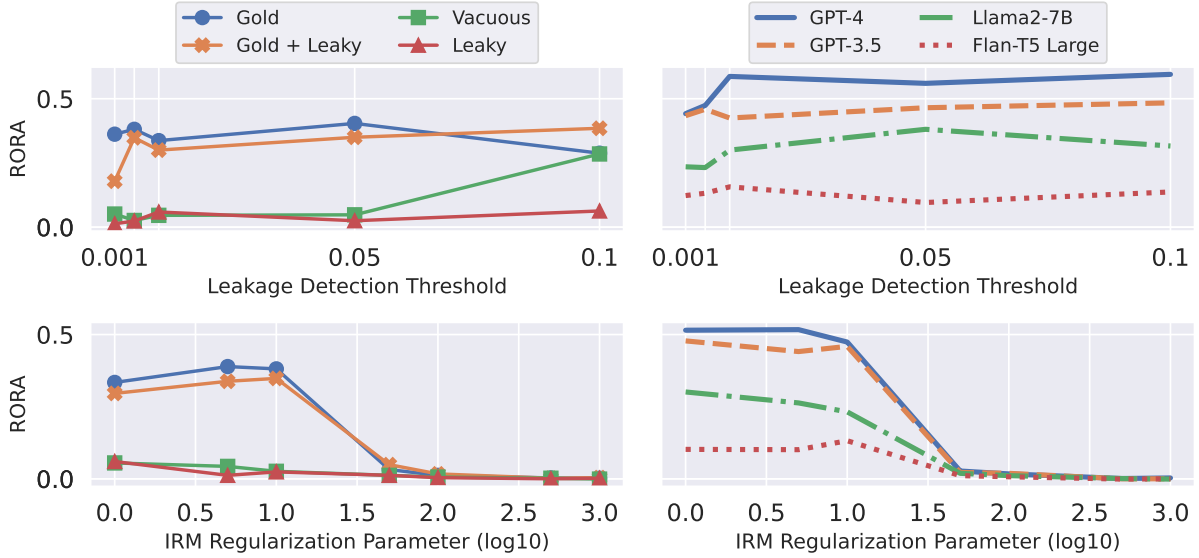


Figure 3: Sensitivity test results of RORA on leakage detection threshold and IRM regularization parameter. **Decreasing threshold and increasing IRM regularization parameter help RORA to better counteract label leakage. RORA appears to be stable when the parameter was chosen from a relatively wide range.**

| Method                  | Pearson's $r$ | Spearman's $\rho$ |
|-------------------------|---------------|-------------------|
| REV (Chen et al., 2021) | 0.05          | 0.03              |
| RORA (Ours)             | 0.36          | 0.37              |

Table 3: RORA achieves a higher correlation with instance-wise human judgments.

notated three-way redundant, and we average all human evaluation scores under the same model to obtain the final score for that particular rationale generation model. Additional human evaluation information is provided in Appendix C.3. The results are shown in Table 2.

While RQ and RORA<sub>ablation</sub> also produce similar rankings as humans that give higher scores to larger LLMs, RORA and RORA<sub>DeBERTa</sub> exhibit closer alignment with human evaluation and provide more fine-grained results. For instance, the RORA score of GPT-3.5 is slightly lower than GPT-4 (0.283  $\rightarrow$  0.253, 0.474  $\rightarrow$  0.459), similar to human evaluation results (2.69  $\rightarrow$  2.32). Furthermore, pointwise  $\mathcal{V}$ -information estimated by RORA for each rationale also has a decent correlation with human judgments, as shown in Figure 2 and Table 3.

## 4.2 Sensitivity Analysis

Considering that RORA first detects leakage tokens based on a predetermined threshold and then leverages IRM to force the evaluation model to debiasing the label leakage, we study RORA's sen-

sitivity towards the choice of the masking threshold as well as the IRM regularization parameter  $\lambda$  in Equation 7.

**Masking Threshold** We test RORA<sub>DeBERTa</sub> across five masking thresholds on StrategyQA, where a lower threshold results in more aggressive detection. Results shown in Figure 3 (first row) indicate that decreasing the threshold does induce stronger intervention, making the model ignorant of label leakage. For instance, the RORA score on *vacuous* rationale drastically declines to 0 when the threshold decreases from 0.1 to 0.01. The *gold<sub>leaky</sub>* rationale is initially higher than *gold*, but after detecting more leakage tokens, it falls slightly below *gold* and keeps having the same trend. Additionally, RORA consistently provides an expected ranking for model-generated rationales across different thresholds, demonstrating strong robustness for free-text rationale evaluation. As discussed in 3.2, we observe that RORA is relatively robust to the threshold because most accidental over-masking is likely to be filled with identical tokens, provided that the token does not have a strong correlation with the label.

**IRM Coefficient  $\lambda$**  The IRM regularization coefficient controls the strength of the invariant learning. A higher value forces the evaluation model to focus more on the shared features between the leaked and non-leaked environments while potentially sacrificing performance to the single environ-

ment. We test  $\text{RoRA}_{\text{DeBERTa}}$  over seven different regularization parameters ranging from 1 to 1000 on StrategyQA and results are shown in Figure 3 (second row). The increase of parameters from 1 to 20 leads to the convergence of lines *gold* and *gold<sub>leaky</sub>*, while still being higher than *vacuous* and *leaky*. This demonstrates that invariant learning is helpful for evaluation models to mitigate label leakage in rationales. However, exceedingly large regularization parameters, such as those higher than 100, can make evaluation models unlearn features in leaked and non-leaked environments, resulting in declined scores close to 0. We also observe that RORA provides a correct ranking for model-generated rationales over a wide and reasonable range of regularization parameters.

### 4.3 Open-Label QA Task

We further test RORA on CommonsenseQA (Tal-  
mor et al., 2019), where we evaluate both ECQA (Aggarwal et al., 2021) and COS-E v1.11 (Rajani et al., 2019). CommonsenseQA adopts a multiple-choice setting where the label set for each problem differs from one another. The model usually needs contextualization to make meaningful predictions when the inputs are linearized into plain text. To have a minimally contextualized attribution model, we use LSTM (Hochreiter and Schmidhuber, 1997) bi-encoder model to separately encode the question and choices and select the final label by calculating the dot product. We perturb ECQA rationales to generate *gold<sub>leaky</sub>*, and other leaky rationales are created similarly as in the previous task.

Table 4 shows that RORA again exhibits strong robustness to label leakages. It ranks *gold* and *gold<sub>leaky</sub>* rationales the highest and gives *vacuous* and *leaky* fairly low scores, while existing baselines are hard to achieve. Additionally, RORA also finds that the quality of CoS-E is inferior to that of ECQA, consistent with the evaluation results from other baselines (Chen et al., 2023b) and human observations (Aggarwal et al., 2021). While REV (Chen et al., 2023b) also ranks all synthetic rationales similarly to RORA, it is because it only takes the rationale as input, and on an open-label dataset, this singlehandedly does not deterministically leak the label. However, our further experiments reveal other potential problems with REV (Appendix B). On the contrary, RORA is not susceptible to such adversarial, as RORA evaluator already has access to everything in the question inputs.

| Rationales | Human-Annotated |             | Synthetic Leaky             |                |              |
|------------|-----------------|-------------|-----------------------------|----------------|--------------|
|            | CoS-E           | <i>gold</i> | <i>gold<sub>leaky</sub></i> | <i>vacuous</i> | <i>leaky</i> |
| RoRA       | 0.398           | 1.30        | 1.33                        | 0.325          | 0.025        |
| REV        | 0.128           | 0.284       | 0.271                       | 0.013          | -0.010       |
| LAS        | 0.198           | 0.365       | 0.373                       | 0.425          | 0.506        |
| RQ         | 0.235           | 0.425       | 0.432                       | 0.402          | 0.432        |

Table 4: Evaluation results on ECQA (Aggarwal et al., 2021) over different rationales, including human-annotated rationales from CoS-E (Rajani et al., 2019). **RORA shows strong robustness in evaluating leaky rationales for open-label QA questions.**

## 5 Related Work

### 5.1 Gradient-based Attribution

Gradient-based attribution methods have been very popular in interpreting neural predictions. Comparative to perturbation (Zeiler and Fergus, 2014; Zintgraf et al., 2017) or Shapley-value-based attributions (Shapley, 1988; Lundberg and Lee, 2017; Strumbelj and Kononenko, 2010; Chen et al., 2020a), variants of gradient-based attribution (Sundararajan et al., 2017; Shrikumar et al., 2017) can be very efficient to compute as the number of samples needed to generate attribution does not depend on the number of features (Ancona et al., 2018; Lyu et al., 2024). Therefore, in this work, we use integrated gradient (IG) (Sundararajan et al., 2017) attribution for the small and small model for label leakage identification because IG is a generally applicable method that has been shown to provide good explanations for NLP tasks (Pruthi et al., 2022), while more easily accessible methods are either non-applicable due to non-linearity being employed in the computation process (Ancona et al., 2018), or requiring specific model structures while still generating questionable attributions (Jain and Wallace, 2019; Wiegrefe and Pinter, 2019; Pruthi et al., 2019; Bibal et al., 2022; Ethayarajh and Jurafsky, 2021; Liu et al., 2022).

### 5.2 Contrastive Generation

To better understand model decision boundary, Gardner et al. (2020) propose to evaluate models with contrast sets, which consist of test instances perturbed in small but meaningful ways that change the gold label. As human supervision in terms of contrast sets helps evaluate and improve classifiers (Kaushik et al., 2019; Chen et al., 2023d), a couple of methods have been proposed to automatically generate counterfactuals as a form of

explanation (Li et al., 2020; Yang et al., 2020; Wu et al., 2021; Dixit et al., 2022; Ross et al., 2022; Jacovi et al., 2021). In particular, SMG (Sha et al., 2021), MICE (Ross et al., 2021), and CREST (Treviño et al., 2023) adopt a select-and-edit two-step process to generate label-changing counterfactuals. In this work, we adapt MICE to generate partial counterfactuals, where only label-leaking parts of a rationale that spuriously correlate to the label get edited.

### 5.3 Invariant Learning

Invariant Risk Minimization (IRM) (Arjovsky et al., 2020; Ahuja et al., 2020) is a recently proposed framework for learning invariant predictors to spurious correlations. While most of these works primarily focused on theoretical findings, simple models, and toy datasets, in a notable departure Drunker et al. (2021) first tests IRM on natural language inference, pointing out that IRM has a potential edge over empirical risk minimization on tasks or datasets with stronger and more prevalent biases and larger data size. This work combines IRM with counterfactual data augmentation to train an evaluator agnostic of label leaking features.

## 6 Conclusions

We introduce RORA, an innovative approach designed to enhance the accuracy and robustness of rationale quality evaluation amidst the challenges posed by label leakage. Our method stands out for its adaptability across diverse model architectures and hyperparameter configurations, delivering consistent rationale quality evaluations that align more closely with human evaluation.

### Limitations

While RORA addresses the problem of label leakage, we focus on the evaluation of the informativeness of a rationale. While an easy-to-follow reasoning path in the rationale seems to boost evaluation score, RORA is not sensitive to other aspects of rationale quality. For example, a good rationale should also be factual (Prasad et al., 2023) and consistent (Chen et al., 2023a), which is not directly addressed in this work.

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# Supplemental Material

| Appendix                   | Contents                                        |
|----------------------------|-------------------------------------------------|
| <a href="#">Appendix A</a> | Justifying Partial Counterfactual Augmentation  |
| <a href="#">Appendix B</a> | Further Discussion on REV and Vacuous Rationale |
| <a href="#">Appendix C</a> | Additional Details of Experimental Setup        |

## A Justifying Partial Counterfactual Augmentation

This section provides intuitive explanations of why other simple alternatives may not work. We consider the following alternatives:

- Mask out all the tokens attributed as leakage by the smaller model.
- IRM training with the environment  $D$  and  $D^{\text{Mask}}$ , which we denote by  $e_1$  and  $e_2$  respectively.

### A.1 Training on Masked Inputs

The major problem is that masking may introduce new artifacts. For instance, in English, negation is applied to the auxiliary verb. Therefore, if the model identifies a discrepancy in tense and a masked token preceding the main verb, it’s highly probable that the masked token represents negation. In our pilot experiments, we found that models are capable of utilizing these traits. Also, attribution calculated from positive and negative labels might be different, a pattern the model might learn to exploit. That’s why in RORA, the attribution is calculated with the global weighting. Partially counterfactual generation makes the sentence more natural, which helps reduce artifacts introduced.

### A.2 Using Masked Inputs as an Environment

We hereby show that invariant learning with a masked environment does not force the model to unlearn  $R_L$ , as described in Equation 5. In general, creating scenarios where one environment includes a non-essential characteristic and the other does not will not result in an Invariant Risk Minimization (IRM) model that achieves comparable loss levels regardless of whether these non-essential characteristics exist. To demonstrate this, we create a minimal example from the causal graph from LAS (Hase et al., 2020), where each simulation variable is replaced with its actual variable as shown in Figure 4.

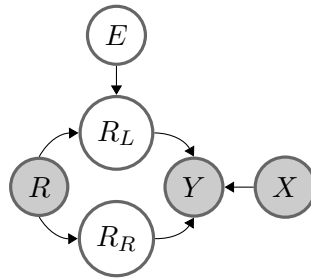


Figure 4: A causal graph showcasing the generative story of the label  $Y$  of an instance, where correctness indicator variables from LAS (Hase et al., 2020) are replaced by an actual variable.

Each variable corresponds to its designation in §2.2. For simplicity, let us consider the following semantics of each random variable:  $R$  is a free-text rationale taking values from all possible rationale documents;  $R_L$  is a ternary variable taking values from  $r_l \in \{\text{True}, \text{False}, \text{UNK}\}$ , where UNK denotes masks;  $R_R$  is a binary indicator of  $r_r \in \{0, 1\}$ ; and  $X$  is the binary indicator of  $x \in \{0, 1\}$  indicating the binary choice innate to the original question. We assume the following conditional probability:

$$\begin{aligned}
p(Y = \text{True} \mid R_L = \text{True}, X, R_R) &= 1, \\
p(Y = \text{True} \mid R_L = \text{False}, X, R_R) &= 0, \\
p(Y = \text{True} \mid R_L = \text{UNK}, X = R_R) &= 0.9, \\
p(Y = \text{True} \mid R_L = \text{UNK}, X \neq R_R) &= 0.
\end{aligned}$$

In this case, the relationship  $R_L \Rightarrow Y$  is *causal*, in the sense that the model can discern which environment the prediction is made in by looking at  $R_L$ . Specifically, assuming that we can identify the leakage part correctly with the application of IG (Sundararajan et al., 2017), then the environments will be

$$\begin{aligned}
e_1 &:= \left\{ (R^{e_1}, X^{e_1}, Y^{e_1}) \mid p(R_L^{e_1} = \text{UNK} \mid R^{e_1}) = 1 \right\}, \\
e_2 &:= \left\{ (R^{e_2}, X^{e_2}, Y^{e_2}) \mid p(R_L^{e_2} = \text{UNK} \mid R^{e_2}) = 1 \right\}.
\end{aligned}$$

Now suppose we have infinite computation, and it is possible to learn any representation  $\Phi : \mathcal{R} \times \mathcal{X} \rightarrow \mathbb{R}^d$ , and a linear classification head  $\tilde{w} : \mathbb{R}^d \rightarrow \mathcal{P}(\mathcal{Y})$  (which outputs a single value between  $[0, 1]$ ). We are considering the risk function  $F$  that is common in classification settings:

$$F(\tilde{w}, \Phi) = \mathbb{E} \left[ -\mathbb{I}[y = \text{True}] \cdot \log(\tilde{w} \circ \Phi(x)) - \mathbb{I}[y = \text{False}] \cdot \log(1 - \tilde{w} \circ \Phi(x)) \right].$$

Suppose we have a representation that fully recovers all the causal components

$$\Phi(r, x) = \left\langle \mathbb{I}[x = q] \cdot \mathbb{I}[r_l = \text{UNK}], \mathbb{I}[r_l = \text{True}] \right\rangle,$$

Then it is easy to see that  $\tilde{w}(z) = v^t z$  where  $v = \langle 0.9, 1 \rangle$  will be the optimal classifier across both environments  $e \in \{e_1, e_2\}$ . It can be verified that

$$\lim_{v' \rightarrow v} \nabla_{w|w=1.0} F(\tilde{w}, \Phi) = -p(Y = \text{True}) \frac{1}{v^T \Phi(x)} + p(Y = \text{False}) \frac{1}{1 - v^T \Phi(x)}.$$

Which, given our  $\tilde{w}$  and  $\Phi$ , does not penalize the predictions. But does this invariant predictor satisfy our desirable property that it will rank instances disregarding whether there is a label leakage? The answer is no, as here, the instances with label leakage will get zero losses, while those without will get non-zero losses. This highlights the importance of generating a counterfactual environment with natural-looking data with a different label association  $p(Y \mid R_L)$ , like what we did for RORA.

## B Further Discussion on REV and Vacuous Rationale

REV (Chen et al., 2023b) evaluates a rationale based on a vacuous rationale  $B$ . We argue that it does not follow the natural way of how humans evaluate a rationale, where we typically need to read and understand the question before the evaluation. Therefore, an adjusted version of REV would be

$$\text{REV}'(R) = I_V(X, R \rightarrow Y \mid B).$$

However, the compositionality nature of language (Pelletier, 1994) implies that in many cases

$$\begin{aligned}
I_V(X, R \rightarrow Y \mid B) - I_V(R \rightarrow Y \mid B) &\neq \\
I_V(X, R \rightarrow Y \mid \emptyset) - I_V(R \rightarrow Y \mid \emptyset). &
\end{aligned} \tag{10}$$

This reveals another potential weakness of REV, which makes restating the question an incredibly high-scoring rationale (0.436 according to our experiment). Indeed, this is counter-intuitive as human evaluators should always have access to  $X$  but not  $B$ . On the contrary, RORA does not have this issue as it already conditions on  $X$ .

## C Experiment Setup

### C.1 Models and Datasets

We access all models through Huggingface Transformers (Wolf et al., 2019). We split the StrategyQA (Geva et al., 2021) train set into our custom train, validation, and test sets and then create model-generated rationales for each set. For ECQA (Aggarwal et al., 2021) we use their original splits.

### C.2 Training Hyperparameters

We train each small model for gradient attribution calculation with a learning rate of  $1e-1$  and batch size of 256. For leakage token detection, we choose threshold 0.005 for  $\text{RORA}_{\text{DeBERTa}}$  and 0.01 for  $\text{RORA}$ , or use top-1 strategy. To determine the appropriate threshold, we assess the output on a small sample of training data, ensuring that at least the majority of the samples receive more than one masking beyond that from the top-k selection. We train each counterfactual data generator using a learning rate of  $1e-4$  with batch sizes 8. For invariant learning, we use IRM regularization  $\lambda = 10$  for both  $\text{RORA}$  and  $\text{RORA}_{\text{DeBERTa}}$  and train each evaluation model with a learning rate of  $1e-4$  and batch size of 64. For each training, we use optimizer AdamW (Loshchilov and Hutter, 2019) from Transformers (Wolf et al., 2019) and set maximum epochs as 20 with early stopping.

### C.3 Human Evaluation Details

We conduct a human evaluation of model-generated rationale on Amazon Mechanical Turk<sup>10</sup>. The quality of each rationale is judged by three individual annotators, who must pass a qualification test described in (Chen et al., 2020b). Annotators have been paid \$0.05 per rationale, which amounts to an hourly wage of about \$10. Figure 5 shows an example of our annotation interface. For each rationale, We present annotators with a question, an answer, and the model-generated rationale (explanation) and ask them the following questions:

1. *Does the Explanation justify the given Answer to the Question?* We offer two options for them to choose from: “Yes” and “No”.
2. If they answer “Yes”, we further ask them *how much additional information does the Explanation have to justify the Answer beyond just reiterating what is stated in Question and Answer?* and to pick one option from “No Additional Info”, “Little Additional Info”, “Some Additional Info” and “Sufficient Additional Info” which are corresponding to a 4-point Likert-scale (0/1/2/3). If the annotator answers “No” for the first question, we label the rationale a score of -1.

Then, we average all three annotators’ scores to get the human evaluation score for that single rationale. For detailed instructions, readers can refer to Figure 6.

### C.4 Model Generated Rationales

Table 6 shows examples of rationale generated by different models.

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<sup>10</sup><https://www.mturk.com/>

**Question:** Is Shakespeare famous becaues of the infinitive form?

**Answer:** The statement is true.

**Explanation:** Shakespeare wrote the play Hamlet. Hamlet contains one of Shakespeare's most famous passages, Hamlet's soliloquy. Hamlet's soliloquy begins with the line 'To be or not to be', which uses the infinitive form.

**Q1:** Does the **Explanation** justify the given **Answer** to the **Question**?

YES

NO

**Q2:** If yes, how much additional information does the **Explanation** have to justify the **Answer** beyond just reiterating what is stated in **Question** and **Answer**?

NO additional info

LITTLE additional info

SOME additional info

SUFFICIENT additional info

SUBMIT

Figure 5: The interface to our human validation hit.

Thanks for participating in this HIT qualifier! Please read the examples below and answer the questions (1-2 questions) to the best of your ability.

**Main Instructions:** you will read a question that usually requires a combination of knowledge about different entities to solve. For each question, an answer and a statement explaining the answer will be provided.

Your task is to read the instance and judge the explanation quality (1-2 questions). Regarding quality, you need to assess two aspects: (1) whether the explanation **supports the label**; (2) whether the explanation contains **additional information** for describing the answer beyond simply combining the question and the answer.

To be more specific:

- **Supports the label** means the explanation is describing something related to the answer to the question (e.g., Example #2 and #3 below), rather than something else (e.g., Example #1 below), and reading it increase your belief in the correctness of the answer.
- **Additional information** means that the explanation provides additional evidence or background knowledge to support the answer (e.g., Example #3 below), rather than a simple paraphrase of the answer (e.g., Example #2 below). You only evaluate the **additional information** when you agree that the explanation **supports the label** first. Please select the **amount of additional information** (from "No additional info" to "Enough additional info") based on the extent to which the explanation helps you understand why the answer might have been chosen for the question.

An instance contains 3 parts:

|                    |                                                                                   |
|--------------------|-----------------------------------------------------------------------------------|
| <b>Question</b>    | A question such as "Is Shakespeare famous because of the infinitive form?"        |
| <b>Answer</b>      | An answer to the question as "The statement is true." (may or may not be correct) |
| <b>Explanation</b> | A <b>statement</b> which explains the <b>Answer</b>                               |

**Notes:**

- **Important!** The **Answer** could be wrong based on your real world knowledge, and the **Explanation** might contain grammatical mistakes. Please **ignore** these when answering the questions.
- You should judge whether the **Explanation** is describing the **Answer** to the **Question** and contains additional information to help you understand why the **Answer** is correct to the **Question**.
- You might see multiple HITs with the same **Question** and **Answer** but different **Explanations**. It's okay. Please evaluate each HIT independently.

Figure 6: The detailed instruction we give the annotators. Notice that we specifically instructed them to not judge the factuality based on their real-world knowledge.

| Dataset | Question                                                                                                   | gold                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | leaky                         | gold <sub>leaky</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | vacuous                            |
|---------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| ECQA    | What town is known for disco? Options: op_1 rio de janeiro op_2 new york op_3 town op_4 germany op_5 hotel | Rio de Janeiro is a huge seaside city in brazil, famed for its Copacabana and Ipanema beaches. The town Rio de Janeiro is known for disco. New York city is famous for Central park, Statue of liberty, Empire state building and other such places and not for disco. Town is larger than a village and generally smaller than a city. Town is not the name of a place where disco is famous for. Germany is the country and not a town known for disco. Hotel is not the town known for disco. | The answer is Rio de Janeiro. | Rio de Janeiro is a huge seaside city in brazil, famed for its Copacabana and Ipanema beaches. The town Rio de Janeiro is known for disco. New York city is famous for Central park, Statue of liberty, Empire state building and other such places and not for disco. Town is larger than a village and generally smaller than a city. Town is not the name of a place where disco is famous for. Germany is the country and not a town known for disco. Hotel is not the town known for disco. The answer is Rio de Janeiro. | Rio de Janeiro is known for disco. |

Table 5: Example of human-annotated and synthetic leaky rationales for open-label QA task. *leaky* verbatim leaks the label, *vacuous* is the declarative combination of the question and the given answer, and *gold<sub>leaky</sub>* is simply the concatenation of the *gold* and *leaky* rationales.

| Model         | Question and Label                                                                            | Generated Rationale                                                                                                                                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPT-4         | Question: Could the Great Wall of China connect the Dodgers to the White Sox?<br>Answer: True | The Great Wall of China is 1,948 km long. The Yankees are based in New York City. The Dodgers are in Los Angeles and play in Los Angeles. The White Sox are based in Chicago and play in Chicago. The answer: yes.                           |
| GPT-3.5       | Question: Could the Great Wall of China connect the Dodgers to the White Sox?<br>Answer: True | No rationale can be provided to explain this answer because it is not accurate. The Great Wall of China is located in China and does not connect the Dodgers (a baseball team in Los Angeles) to the White Sox (a baseball team in Chicago). |
| Llama2-7B     | Question: Could the Great Wall of China connect the Dodgers to the White Sox?<br>Answer: True | The Great Wall of China is a long wall. The Dodgers and White Sox are baseball teams.                                                                                                                                                        |
| Flan-T5 Large | Question: Could the Great Wall of China connect the Dodgers to the White Sox?<br>Answer: True | The Great Wall of China is 1,948 km long. The Yankees are based in New York City. The Dodgers are in Los Angeles and play in Los Angeles. The White Sox are based in Chicago and play in Chicago. The answer: yes.                           |

Table 6: Examples of model-generated rationales